

ECE 222 In-Class Problems 1

In this discussion, we want to build your intuition for and comfort with vector fields and vector calculus. Discuss these problems with your neighbors, write down your answers, then discuss as a class. Focus on the “why”, not just the correct answer.

At the end of class, take a photograph(s) or screenshot(s) of your answers and upload to today’s in-class problem assignment on Canvas.

Problem 1:

Repeat homework problem 3.44 a, b, d, e, g, and h (below), but:

- FIRST, just look (no math) at the vector map and answer these questions:
 - Is the field solenoidal (divergenceless)? If not, where is it positive and where is it negative?
 - Is the field conservative (irrotational or no curl everywhere)? If not, where does it point in + and where does it point in – ?
 - It may help to make sketches of the maps and indicate where you expect the field to be solenoidal, conservative, or have positive or negative divergence or curl.
- SECOND, check a few of your answers (as you have time) by doing the math to find the divergence and curl.

3.44 Each of the following vector fields is displayed in Fig. P3.44 in the form of a vector representation. Determine $\nabla \cdot \mathbf{A}$ analytically and then compare the result with your expectations on the basis of the displayed pattern.

a. $\mathbf{A} = -\hat{x} \cos x \sin y + \hat{y} \sin x \cos y$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(a)

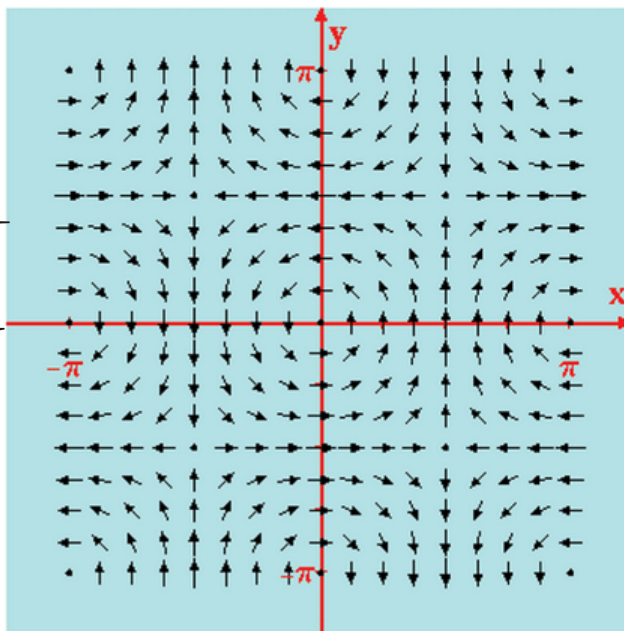
$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} = \sin x \sin y - \sin x \sin y = 0$$

$$\nabla \times \mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$(\cos x \cos y - (-\cos x \cos y)) = 2 \cos x \cos y$$

So: Solenoidal & not conservative

Curl + when $2 \cos x \cos y \hat{z}$ is + and - respectively; signs flip @ $xy = \pm \frac{\pi}{2}$



By looking at it, there is curl > 0 at (0,0) about $\hat{z}$; it looks like there is no $\text{div} \neq 0$ anywhere...

b. $\mathbf{A} = -\hat{x} \sin 2y + \hat{y} \cos 2x$, for $-\pi \leq x, y \leq \pi$

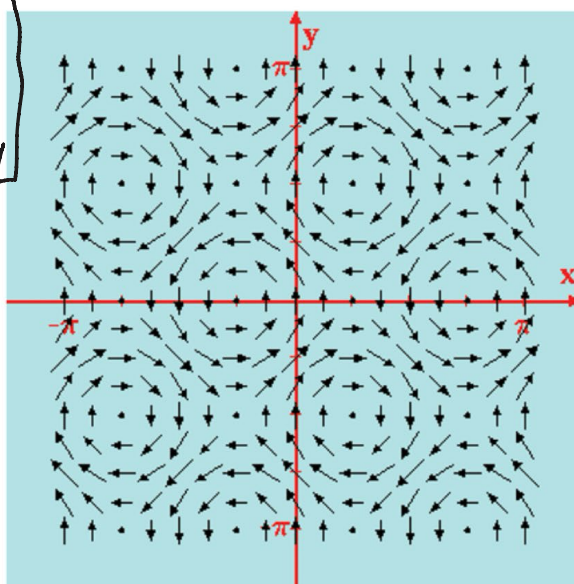
Figure P3.44(b)

$$\nabla \cdot \mathbf{A} = 0 + 0$$

$$\nabla \times \mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$-2 \sin 2x + 2 \cos 2y$$

Solenoidal, not conservative



Curl < 0 @ (1,2), (-1,2)...

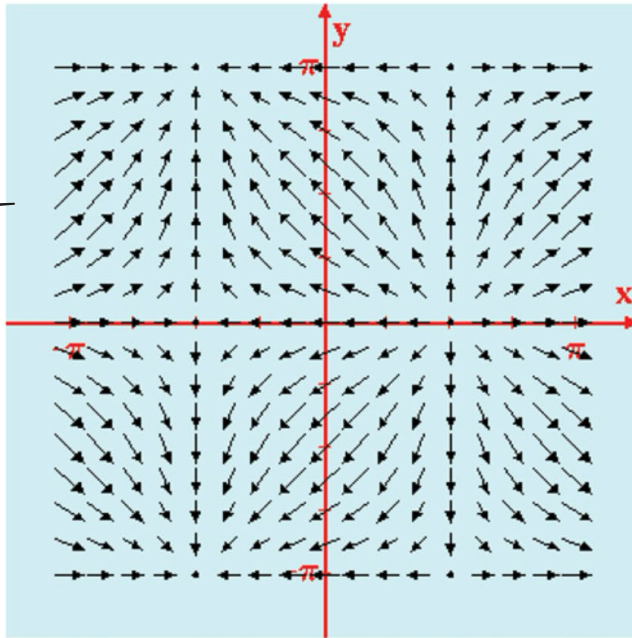
Curl > 0 @ (-2,0), (6,0)...

div looks like zero everywhere...

d. $\mathbf{A} = -\hat{x} \cos x + \hat{y} \sin y$, for $-\pi \leq x, y \leq \pi$

Figure P3.44(d)

$\nabla \cdot \mathbf{A} = \sin x + \cos y$, $\nabla \times \mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$
 Not Solenoidal & Conservative

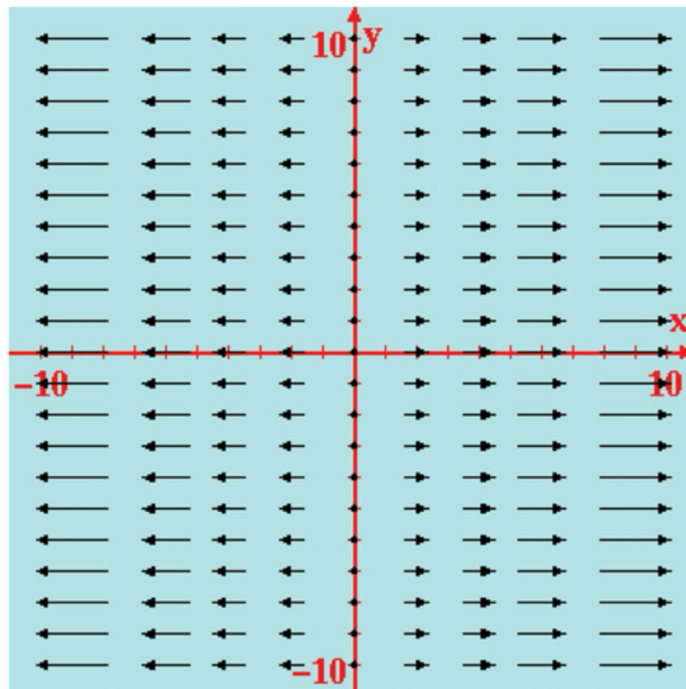


$\text{Div} > 0$ @ (2,0)
 Curl looks like zero everywhere

e. $\mathbf{A} = \hat{x} x$, for $-10 \leq x \leq 10$

Figure P3.44(e)

$\nabla \cdot \mathbf{A} = 1$, $\nabla \times \mathbf{A} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$
 Not Solenoidal but conservative

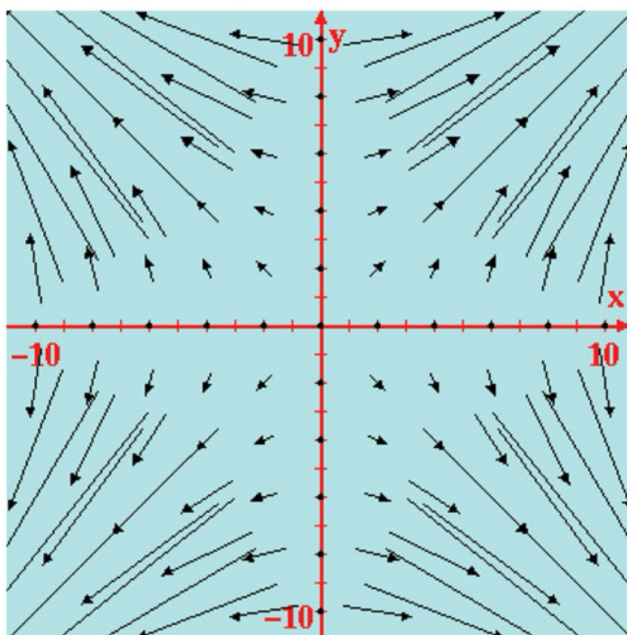


Maybe $\text{div} > 0$?
 curl is zero everywhere it looks.

g. $\mathbf{A} = \hat{x} xy^2 + \hat{y} x^2 y$, for $-10 \leq x, y \leq 10$

Figure P3.44(g)

$\nabla \cdot \mathbf{A} = y^2 + x^2$, $\nabla \times \mathbf{A} = \begin{bmatrix} 0 & -0 \\ 0 & -0 \\ 2xy & -2xy & 0 \end{bmatrix}$
 Not Solenoidal and
 Conservative

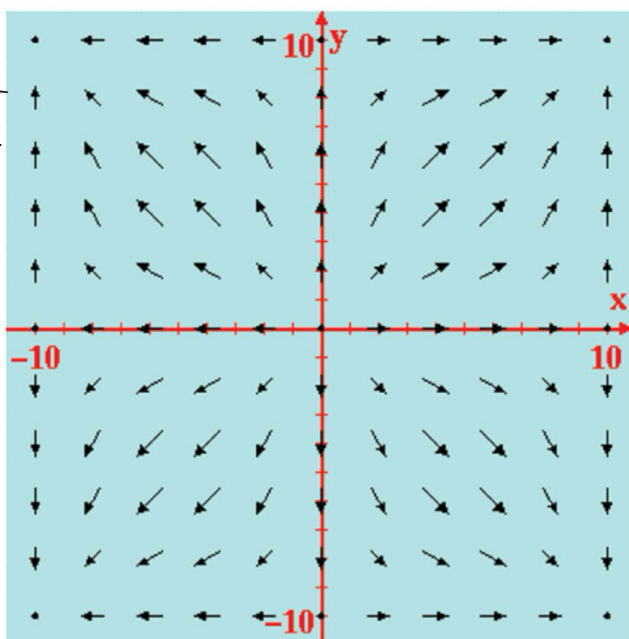


$\text{div} > 0$ @ (3,3)
 it seems
 $\text{curl} = 0$ everywhere
 I think

h. $\mathbf{A} = \hat{x} \sin\left(\frac{\pi x}{10}\right) + \hat{y} \sin\left(\frac{\pi y}{10}\right)$, for $-10 \leq x, y \leq 10$

Figure P3.44(h)

$\nabla \cdot \mathbf{A} = \frac{\pi}{10} \cos\left(\frac{\pi x}{10}\right) + \frac{\pi}{10} \cos\left(\frac{\pi y}{10}\right)$
 $\nabla \times \mathbf{A} = \begin{bmatrix} 0 & -0 \\ 0 & -0 \\ 0 & 0 \end{bmatrix}$
 Not Solenoidal & Conservative



$\text{div} > 0$ @ (3,3)
 Looks conservative.

Problem 2:

Examine homework problem 3.58 below. **Without doing any math**, which of these do you expect to be conservative? **Why?**

If you are having trouble with this problem, determine the curl mathematically for a few of the vector fields, and look at what those fields have in common.

3.58 Determine if each of the following vector fields is solenoidal, conservative, or both:

- * a. $\mathbf{A} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}2xy$
- b. $\mathbf{B} = \hat{\mathbf{x}}x^2 - \hat{\mathbf{y}}y^2 + \hat{\mathbf{z}}2z$
- c. $\mathbf{C} = \hat{\mathbf{r}}(\sin \phi)/r^2 + \hat{\phi}(\cos \phi)/r^2$
- * d. $\mathbf{D} = \hat{\mathbf{R}}/R$
- e. $\mathbf{E} = \hat{\mathbf{r}} \left(3 - \frac{r}{1+r} \right) + \hat{\mathbf{z}}z$
- f. $\mathbf{F} = (\hat{\mathbf{x}}y + \hat{\mathbf{y}}x)/(x^2 + y^2)$
- g. $\mathbf{G} = \hat{\mathbf{x}}(x^2 + z^2) - \hat{\mathbf{y}}(y^2 + x^2) - \hat{\mathbf{z}}(y^2 + z^2)$
- * h. $\mathbf{H} = \hat{\mathbf{R}}(Re^{-R})$

At a glance: B, D, E, H look so because vector terms depend on their own variable.
I'll check C & G:

$$C. \nabla \times C = \begin{bmatrix} \frac{1}{r} \cdot 0 - 0 \\ 0 - 0 \\ \frac{1}{r} \left(\frac{\partial}{\partial r} (\cos \phi / r) - \frac{\cos \phi}{r^2} \right) - \frac{\cos \phi}{r^2} \end{bmatrix} = \frac{1}{r} \left(-\frac{\cos \phi}{r^2} - \frac{\cos \phi}{r^2} \right) = -\frac{2 \cos \phi}{r^3}$$

G: Not conservative

$$\nabla \times G = \begin{bmatrix} -2y - 0 = -2y \\ 2z - 0 = 2z \\ -2x - 0 = -2x \end{bmatrix}$$

Not conservative.